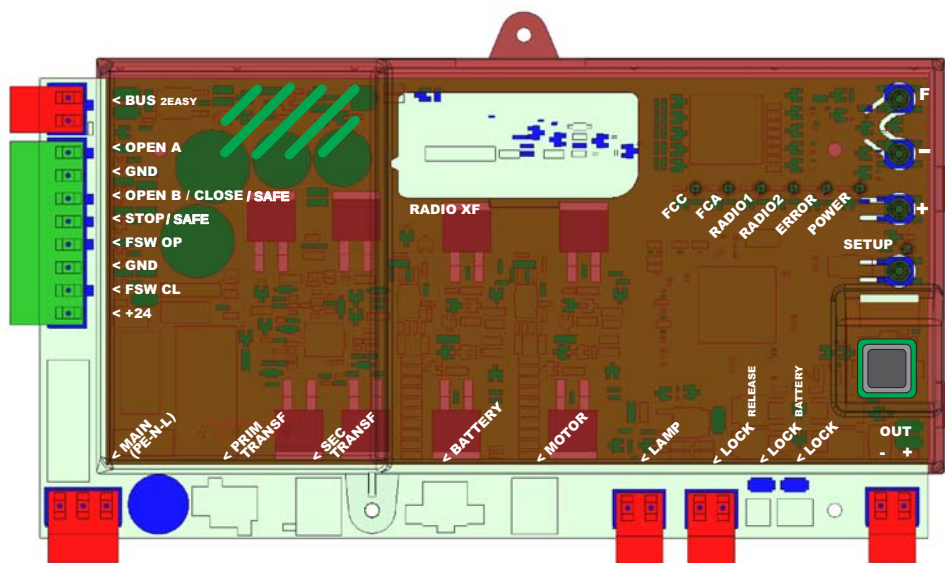


E720



CONTENTS

1 WARNINGS	2
2 LAYOUT AND COMPONENTS	2
2.1 COMPONENT DESCRIPTION	3
2.2 DESCRIPTION OF TERMINAL BLOCK J13.....	3
3 TECHNICAL SPECIFICATIONS	3
4 ELECTRIC CONNECTIONS	4
4.1 BUS-2EASY PHOTOCELLS.....	5
4.1.1 ADDRESSING BUS-2EASY PHOTOCELLS	5
4.1.2 STORING BUS-2EASY ACCESSORIES	5
4.2 TRADITIONAL PHOTOCELLS	5
5 PROGRAMMING	7
5.1 1 st LEVEL PROGRAMMING	7
5.2 2 nd LEVEL PROGRAMMING	9
6 STORING RADIO CODE	11
6.1 STORING DS RADIOCONTROLS	11
6.2 STORING SLH RADIOCONTROLS	12
6.3 STORING LC/RC RADIOCONTROLS (ONLY FOR SOME MARKETS)	12
6.3.1 STORING LC/RC RADIOCONTROLS REMOTELY	12
6.4 RADIOCONTROLS DELETION PROCEDURE	12
7 OPERATIONAL START-UP	13
7.1 CHECKING THE LEDS	13
7.2 POSITIONING LIMIT SWITCHES	14
7.3 SETUP	15
8 TESTING THE AUTOMATED SYSTEM	15
9 ALARM AND ERROR SIGNALS	15
9.1 ALARMS	15
9.2 ERRORS	15
10 FUNCTION LOGICS	16

EC DECLARATION OF CONFORMITY

Manufacturer: FAAC S.p.A.

Address: Via Calari, 10 - 40069 Zola Predosa BOLOGNA - ITALY

Declares that: Electronic device E720

- conforms to the essential safety requirements of the following EEC directives

2006/95/EC Low Voltage Directive

2004/108/EC Electromagnetic Compatibility Directive

Additional note:

This product has undergone testing in a typical standard configuration (all products built by FAAC S.p.A.)

Bologna, 01-05-2010

The Managing Director
A. Marcellan



WARNINGS

- Attention! To ensure the safety of people, it is important that you read all the following instructions.
- Incorrect installation or incorrect use of the product could cause serious harm to people.
- Carefully read the instructions before beginning to install the product and keep for future reference.
- The symbol  highlights notes that are important for personal safety and the protection of the automated system.
- The symbol  calls your attention to notes on product specifications or operation.

E720

1 WARNINGS



Before attempting any work on the electronic device (connections, maintenance), always turn off power.



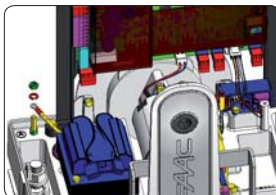
Install, upstream of the system, a differential thermal breaker with adequate tripping threshold.



Connect the earth lead to the appropriate terminal.



Always separate power cables from control and safety cables (push-button, receiver, photocells, etc.). To avoid any electric noise, use separate sheaths or a shielded cable (with earthed shield).



2 LAYOUT AND COMPONENTS

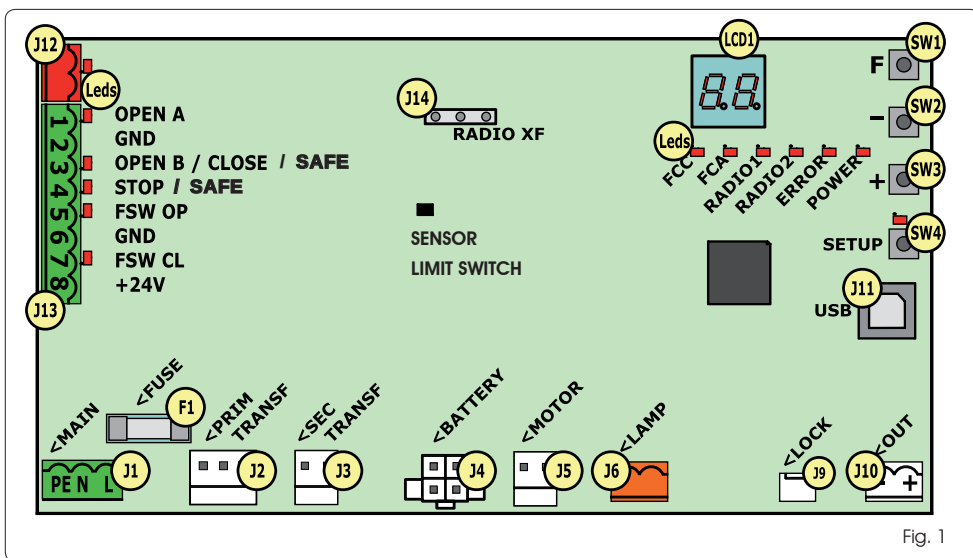


Fig. 1

2.1 COMPONENT DESCRIPTION

LCD	SIGNALLING AND PROGRAMMING DISPLAY
SW1	PROGRAMMING PUSH-BUTTON "F".
SW2	PROGRAMMING PUSH-BUTTON "-".
SW3	PROGRAMMING PUSH-BUTTON "+".
SW4	"SETUP" PUSH-BUTTON
Leds	INPUTS STATUS CONTROL LED
J1	MAIN POWER SUPPLY CONNECTOR
J2	TRANSFORMER PRIMARY WINDING CONNECTOR
J3	TRANSFORMER SECONDARY WINDING CONNECTOR
J4	EMERGENCY BATTERY CONNECTOR (ACCESSORY)
J5	MOTOR CONNECTOR
J6	FLASHER OUTPUT CONNECTOR
J9	MOTOR LOCK AND MOTOR RELEASE CONTACT CONNECTOR
J10	OUT OUTPUT CONNECTOR
J11	USB CONNECTOR FOR PC CONNECTION
J12	BUS-2EASY DEVICE CONNECTION CONNECTOR
J13	INPUT CONNECTOR IN CONNECTOR BLOCK
J14	RADIO RECEIVER MODULE CONNECTOR FOR OMNIDEC
LCD1	SIGNALLING AND PROGRAMMING DISPLAY
F1	PROTECTION FUSE

2.2 DESCRIPTION OF TERMINAL BLOCK J13

INPUT	No	DESCRIPTION
1	OPEN A	Device with N.O. contact that causes total opening of the gate
2-6	GND	Accessory power supply negative
3	OPEN B / CLOSE / SAFE	Device with N.O. contact that causes partial opening of the gate or reclosing of the gate according to the selected function logic
4	STOP / SAFE	Device with N.C. contact that halts the gate
5	FSW OP	Device with N.C contact that reverses the motion during gate opening
7	FSW CL	Device with N.C contact that reverses the motion during gate closing
8	+24 V	Accessory power supply positive

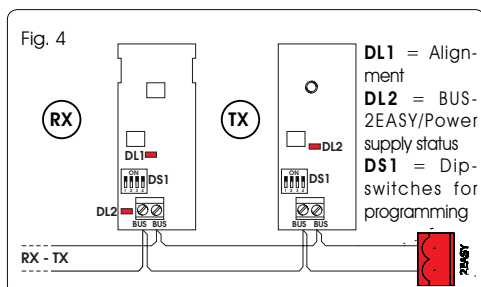
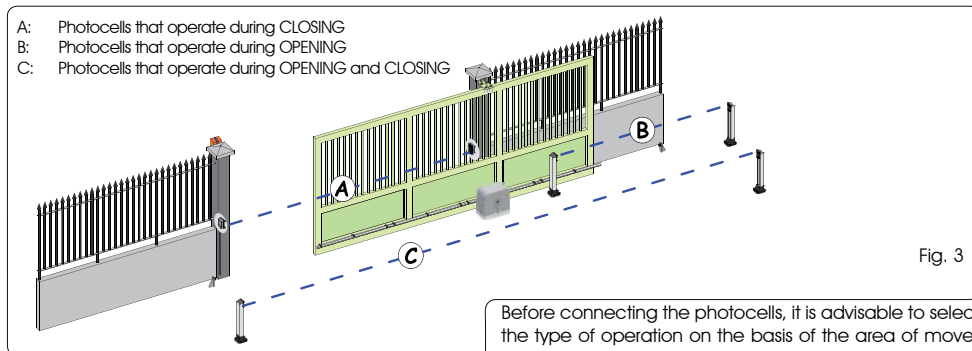
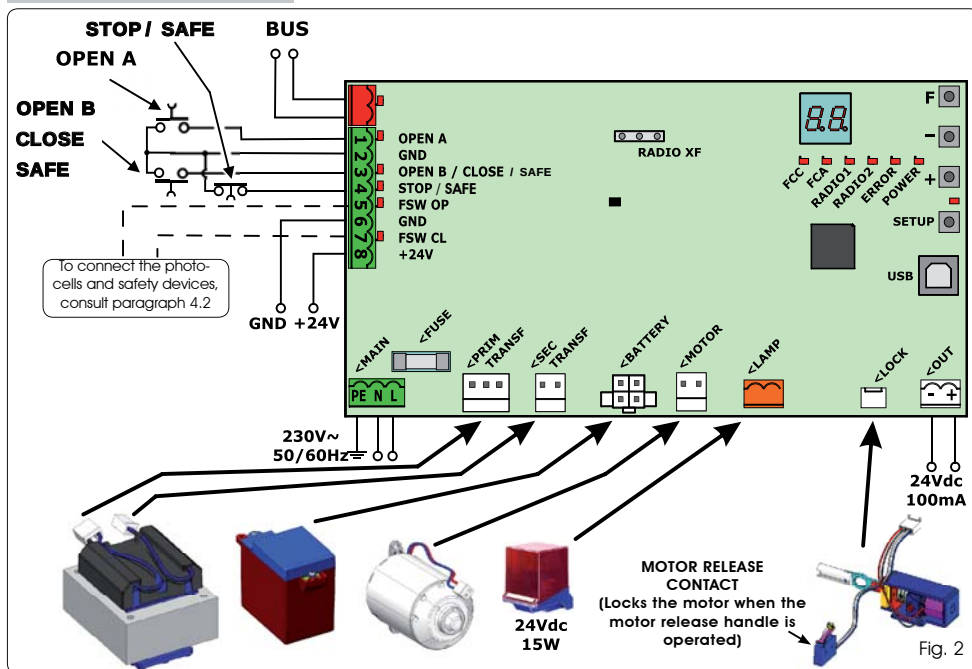
3 TECHNICAL SPECIFICATIONS

Power supply	230 V~ 50 Hz
Power consumption from mains stand-by	10 W
Motor max. load	6 A
Accessory power supply	24 Vdc
Accessory max. current	24 Vdc max. 500 mA BUS-2EASY max. 500 mA
Environmental temperature	(-20 - +55) °C
Flasher load	24 Vdc - 15 W
Output load	24 Vdc - 100 mA (1)
Protection fuses	F1 = 11A - 250V
Function logics	Semiautomatic, Semiautomatic "step", Automatic, Automatic "step", Automatic with timer function, Automatic Safety devices, Automatic Safety devices "step", Automatic with reverse on pause, Semiautomatic "b", Mixed logic "bC", Dead-man.
Work time	Programmable (from 0 to 10 min.)
Pause time OPEN A / OPEN B	Programmable (from 0 to 10 min.)
Motor power	Adjustable over 50 levels
Opening-closing motor speed	Adjustable over 10 levels
Connector Inputs/Outputs	Power supply, Battery, Motor, Module XF433/868, Motor lock electric release batteries, Motor Lock, USB
Inputs/Outputs in terminal block	BUS-2EASY, OPEN A, OPEN B/CLOSE/SAFE, STOP/SAFE, GND, Opening and closing photocells, +24 V, Mains power supply, Flasher, Electric release motor lock, OUT
Programming	1 st and 2 nd level with 3 keys (+, -, F) and display.



(1) The output load must be considered as already included in the max. current available for the accessories

4 ELECTRIC CONNECTIONS



Before connecting the photocells, it is advisable to select the type of operation on the basis of the area of movement that they need to protect:

Safety devices during closing: operate only during the automated system closing movement and are therefore suitable for protecting the closing area from the risk of impact.

Safety devices during opening: operate only during the automated system opening movement and are therefore suitable for protecting the opening area from the risk of impact.

Safety devices during opening/closing: operate during both the automated system opening and closing movements and are therefore suitable for protecting the entire movement area from the risk of impact.

4.1 BUS-2EASY PHOTOCELLS

This board is equipped with a BUS-2EASY circuit that can be used to easily connect a high number of auxiliary BUS-2EASY devices to the safety device (e.g. up to 16 pairs of photocells), appropriately programmed, using only two cables without polarity.

Before connecting the photocells, it is advisable to select the type of operation (Fig. 3) on the basis of the area of movement the cells must protect and to position the dip switches on both the transmitter and receiver (see Fig. 4) as in Tab. 1.

4.1.1 ADDRESSING BUS-2EASY PHOTOCELLS

It is important to give both the transmitter and the receiver the same address.

Ensure that there are not two or more photocell pairs with the same address.

If no BUS-2EASY accessory is used, leave the BUS-2EASY connector (J12 - fig. 1) free.

Tab. 1 - Addressing BUS-2EASY photocells

Dip1	Dip2	Dip3	Dip4	Re.	Type
OFF	OFF	OFF	OFF	B	OPENING Max. 6 pairs
OFF	OFF	OFF	ON		
OFF	OFF	ON	OFF		
OFF	OFF	ON	ON		
OFF	ON	ON	OFF		
OFF	ON	ON	ON		
ON	OFF	OFF	OFF	A	CLOSING Max. 7 pairs
ON	OFF	OFF	ON		
ON	OFF	ON	OFF		
ON	OFF	ON	ON		
ON	ON	OFF	OFF		
ON	ON	ON	OFF		
OFF	ON	OFF	OFF	C	OPENING and CLOSING Max. 2 pairs
OFF	ON	OFF	ON		
ON	ON	ON	ON	/	OPEN PULSE

4.1.2 STORING BUS-2EASY ACCESSORIES

It is possible to add BUS-2EASY photocells to the system at any time, simply by following the procedure below:

1. Install and programme the accessories with the required address (see par. 4.1.1).
2. Cut off power to the board.
3. Connect both cables of the BUS-2EASY accessories to the red terminal block J12 (polarity irrelevant).
4. Power the board.
5. Quickly press the SETUP push-button (SW4) once to register the accessories. Check the operation of the installed BUS-2EASY devices.
6. The board has stored the BUS-2EASY accessories.

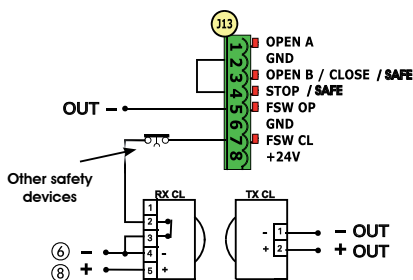
Follow the instructions in the following table to check that the BUS-2EASY connection status is efficient.

Tab. 2 - Description of BUS-2EASY led

Fixed ON	Normal activity (led on even without photocells). No registered photocell engaged.
Slow flasher	At least one registered photocell engaged or not aligned.
Off (flash every 2.5 secs)	BUS-2EASY line short-circuited.
Off	BUS-2EASY line deactivated.

4.2 TRADITIONAL PHOTOCELLS

Connection of 1 pair of closing photocells with FAIL-SAFE safety device activated
Set in second level of programming
 $\square 1 = \square 1$



Connection of 1 pair of closing photocells with FAIL-SAFE and STOP safety device deactivated

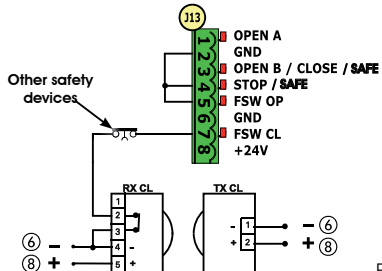


Fig. 5

If the FAIL-SAFE safety device is not used, connect the power supply of the transmitters to terminals 6 and 8 of J13.

If the FAIL-SAFE safety device is used, connect the power supply of the transmitters to OUT after setting it as appropriate (see 2nd level programming and Fig. 16).

If the FAIL-SAFE safety device is used, even the unused safety inputs must be connected via a shunt lead to the negative of OUT (see Fig. 16).

Connection of a pair of opening photocells

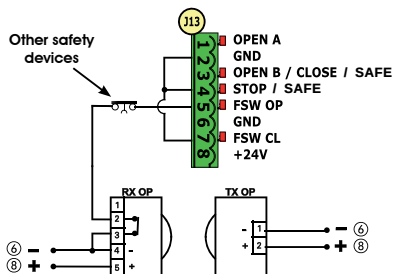


Fig. 6

Connection of a pair of closing photocells and a pair of opening photocells

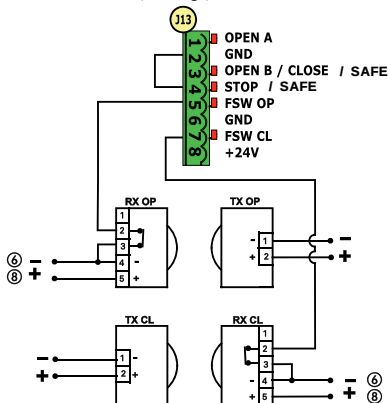


Fig. 7

Connection of two pairs of closing photocells

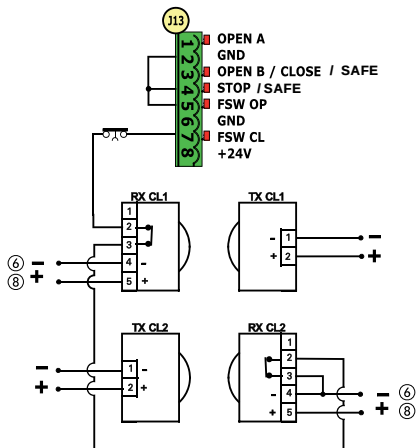


Fig. 8

Connection of a pair of closing photocells and a pair of opening/closing photocells

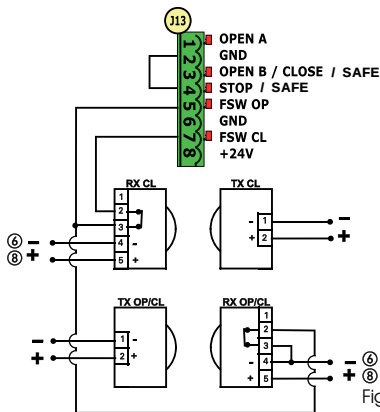


Fig. 9

Connection of a pair of closing photocells, a pair of opening photocells and a pair of opening/closing photocells

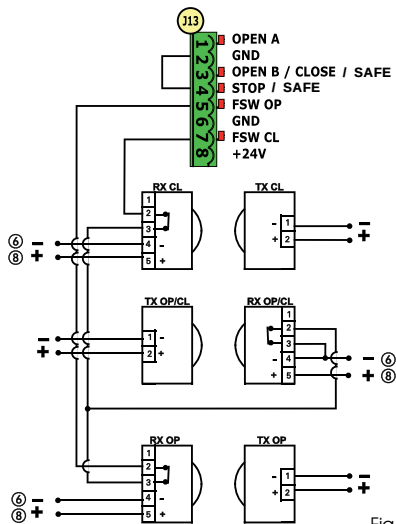


Fig. 10

Connection of no safety and stop device

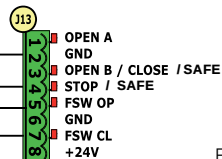


Fig. 11

5 PROGRAMMING

It is necessary to access PROGRAMMING mode in order to program automated system operation.

Programming is divided in two parts: 1st LEVEL and 2nd LEVEL.



The status of the automated system is normally shown on the display. When push-button F is pressed or when F and + are pressed simultaneously, this gives access to 1st or 2nd level programming.



To restore the default settings, simply reload the desired default in the first passage of the 1st level programming.



The change of the programming parameters becomes effective, while final storage will not take place until programming mode has been exited and the input status is again displayed. If power supply to the device is cut before going back to the input status display, all the changes made will be lost.



It is possible to return to the input status display and store all the parameters modified to that moment from any point of the 1st and 2nd level programming by pressing keys F and - simultaneously.



When the board is turned on, the software version of the board as two figures separated by a decimal point appears on the LCD1 display.

5.1 1st LEVEL PROGRAMMING

To access 1st level programming, press push-button F.

- If you press key **F** again (and hold it down), the display shows the name of the function.
- If you release the key, the display shows the value of the function that can be modified with keys + and -.
- If you press **F** again (and hold it down), the display shows the name of the next function, etc.
- When you reach the last function, press push-button **F** to exit the program and store the parameters. The display shows again the status of the automated system.

1 st LEVEL PROGRAMMING 		
Display	Function	Default
df	DEFAULT: I Configures the parameters with DEFAULT values. CU If push-button F is released, the value CU appears, this means that the standard modified configuration has been selected by means of push-buttons and display. If you wish to maintain this programming, press push-button F again, otherwise press + and select default I .	I
LO	FUNCTION LOGICS: EP Semiautomatic. SP Semiautomatic "Step". AI Automatic. PI Automatic 1. AP Automatic "Step". AT Automatic with timer function. AS Automatic "Safety devices". SP Automatic Safety Devices "Step". SA Automatic with reverse during pause. GB Semiautomatic "b". MC Mixed (AP pulse/ CH dead-man). CD Dead-man.	E
PA	PAUSE TIME A: Pause time with a TOTAL opening command. This has effect only if the automatic logic was selected. Adjustable from 0 to 59 secs. in one-second steps. Subsequently, display changes to minutes and tens of seconds (separated by a point) and time is adjusted in 10-second steps, up to the maximum value of 9.5 minutes. E.g. if the display shows 2.5 , pause time is 2 min. and 50 secs.	20
Pb	PAUSE TIME B: Pause time with a PARTIAL opening command. This has effect only if the automatic logic was selected. Adjustable from 0 to 59 secs. in one-second steps. Subsequently, display changes to minutes and tens of seconds (separated by a point) and time is adjusted in 10-second steps, up to the maximum value of 9.5 minutes. E.g. if the display shows 2.5 , pause time is 2 min. and 50 secs.	20

Display	Function	Default
F0	MOTOR POWER: Adjusts the motor power level 01 = minimum power 50 = maximum power  When the power value has been changed, it is advisable to perform a new SETUP (see par. 7.3)	50
S0	OPENING SPEED: Adjusts the motor opening speed to 10 levels 01 = minimum speed 10 = maximum speed  When the speed value has been changed, it is advisable to perform a new SETUP (see par. 7.3)	08
S2	CLOSING SPEED: Adjusts the motor closing speed to 10 levels. 01 = minimum speed 10 = maximum speed  When the speed value has been changed, it is advisable to perform a new SETUP (see par. 7.3)	08
r0	SLOWING DURING OPENING Adjusts the slowing space as a percentage of the total gate travel. Adjustable from 5 (with speed <5), 10 (with speed between 5 and 8), 15 (with speed >8) to 99 % in 1% steps. 5-10-15 = minimum slowing 99 = maximum slowing	20
rc	SLOWING DURING CLOSING: Adjusts the slowing space as a percentage of the total gate travel. Adjustable from 5 (with speed <5), 10 (with speed between 5 and 8), 15 (with speed >8) to 99 % in 1% steps. 5-10-15 = minimum slowing 99 = maximum slowing	20
Sr	SPEED DURING SLOWING: Adjusts gate speed during slowing. 0 = LOW speed 1 = HIGH speed	0
St	STATUS OF AUTOMATED SYSTEM: Exit from programming, data storage, and return to gate status display. 00 = CLOSED 01 = OPEN 02 = Stop then "OPEN" 03 = Stop then "CLOSE" 04 = In "PAUSE" 05 = During opening stage 06 = During closing stage 07 = FAIL SAFE in progress 08 = BUS-2EASY device check in progress 09 = Preflashing then "OPEN" 10 = Preflashing then "CLOSE" 11 = OPEN in Partial Opening 12 = in PAUSE Partial Opening	

5.2 2nd LEVEL PROGRAMMING

To access 2ND LEVEL PROGRAMMING, press push-button **F** and hold down while pressing push-button **+**:

- if you also release push-button **F**, the display shows the value of the function that can be modified with keys **+** and **-**.
- if you press key **F** (and hold down), the display shows the name of the next function. If you release it, the value that can be modified with keys **+** and **-** is displayed.
- when you reach the last function, press **F** to exit programming, and the display resumes showing the automated system status.

2 nd LEVEL PROGRAMMING  + 		
Display	Function	Default
bo	MAXIMUM TAKE-OFF POWER: The motor works at maximum power (ignoring the selected power level) during the movement take-off. y = active no = excluded	no
PF	PREFLASHING: Allows 5 types of preflashing to be selected, with a duration equal to 3 secs. no = no preflashing. OC = preflashing before each movement. CL = preflashing before a closing movement. OP = preflashing before an opening movement. PA = preflashing only at end of pause.	no
tp	PREFLASHING TIME (visible only if preflashing is selected in the previous menu): Adjusts preflashing time from 1 to 10 sec. in 1 second steps. 01 = minimum preflashing 10 = maximum preflashing	03
Ph	CLOSING PHOTOCELLS: Activate the function if you want the closing photocells to lock movement and reverse it on disengagement. Normally, operation of the closing photocells immediately puts the gate into reverse when this function is off. y = reversal on disengagement no = immediate reversal to opening	no
oP	OPENING PHOTOCELLS: Activate the function if you want the opening photocells to lock movement and reverse it during closing. Normally, with this function off, operation of the opening photocells determines the resumption of movement on their disengagement. no = immediate reversal to closing no = resumption of motion on disengagement	no
Ad	ADMAP FUNCTION Permits activation of operation in accordance with French standard NFP 25/362. y = active no = excluded	no
EC	ENCODER: The "virtual" encoder on the board operates as an anti-crushing device and manages the slowing and partial opening start points. If the gate strikes an obstacle during the opening or closing stage, the "virtual" encoder causes reversal of the motion. The second obstacle detected in the same direction causes the motor to stop. An alarm is signalled in both cases (see section on alarms). It is necessary to adjust the sensitivity of the "virtual" encoder by varying the time that the board waits for before controlling reversal of motion from a minimum of 0 sec to a maximum of 10 sec in one-second steps. 00 = maximum sensitivity 10 = minimum sensitivity	02

Display	Function	Default
P0	PARTIAL OPENING: It is possible to adjust the width of the partial leaf opening as a percentage of the total gate travel. Adjustable from 0 to 99% in 1% steps. 00 = no partial opening 01 = minimum partial opening 99 = maximum partial opening	50
t	TIME-OUT It is advisable to set a value higher than the time taken by the gate to open or close fully. Adjustable from 0 to 59 sec. in 1 second steps. Subsequently, display changes to minutes and tens of seconds (separated by a point) and time is adjusted in 10-second steps, up to the maximum value of 9.5 minutes. E.g.: if the display shows 2.5 , pause time is 2 min. and 50 secs.	2.0
01	OUT 1: Allows output OUT (open collector N.O.) to be activated in one of the following functions: 00 = Always active 01 = FAIL-SAFE 02 = WARNING LIGHT (off when closed, on during opening and open/pause, flashing during closing) 03 = COURTESY LIGHT (see next function) 04 = ALARM, BATTERY operation 05 = gate OPEN or PAUSED 06 = gate CLOSED 07 = gate in MOTION 08 = gate OPENING 09 = gate CLOSING 10 = ACTIVE safety 11 = timed output that can be activated by second radio channel (see next function) 12 = output that can be activated by second radio channel (step function)	00
t1	TIMING OF OUT 1 (only if option 03 or 11 is selected at the previous step): Can be used to adjust timing of output OUT if a timed function was selected (e.g. 03 or 11) from 1 to 99 minutes in 1 minute steps.	02
0b	OPEN B / CLOSE / EDGE SAFETY DEVICE: It is possible to select the use of OPEN B input as partial opening, CLOSE command or EDGE SAFETY DEVICE. 00 = OPEN B 01 = CLOSE 02 = EDGE SAFETY DEVICE  <i>If you select a function logic that involves the use of the CLOSE command (logic b, bC, C) this function will be preset to 01 and it will not be possible to change it.</i>	00
SP	STOP / EDGE SAFETY DEVICE It is possible to select the use of STOP input as STOP or EDGE SAFETY DEVICE. 00 = STOP 01 = EDGE SAFETY DEVICE	00
Lc	MOTOR LOCK: It is possible to select the use of the motor lock during battery operation: Y = the motor lock continues to operate regularly even during battery operation no = when the system switches to battery operation, the motor lock remains open.	Y

Display	Function	Default														
AS	SERVICE REQUEST - CYCLE COUNTER (combined with the two successive functions): Y = when the set number of cycles is reached with successive functions nc and nd, the system carries out an 8 sec. preflashing sequence (in addition to the sequence already set using the PF function) before each movement. no = successive functions nc and nd indicate how many cycles the system has performed up to the maximum possible display of 99.990. If the number of cycles is higher than 99.990, the two successive functions nc and nd will show 99 and 99 respectively. This function may be useful for setting programmed maintenance operations or for checking work cycles performed.	no														
nc	CYCLE PROGRAMMING (THOUSANDS): If AS = Y the display indicates the number of thousands of cycles after which service is required (may be set from 0 to 99). If AS = no the display indicates the number of thousands of cycles performed. The value displayed is updated with the sequence of cycles, interacting with the value of nd. If AS = no press keys + and - for 5 sec. to zero the cycle counter.	00														
nd	CYCLE PROGRAMMING (TENS): If AS = Y the display indicates the number of tens of cycles after which service is required (may be set from 0 to 99). If AS = no the display indicates the number of tens of cycles performed. The value displayed is updated with the sequence of cycles, interacting with the value of nc. Example: if the system has performed 11.218, it will display nc = 11 and nd = 21	00														
St	STATUS OF AUTOMATED SYSTEM: Exit from programming, data storage, and return to gate status display. <table><tr><td>00 = CLOSED</td><td>07 = FAIL SAFE in progress</td></tr><tr><td>01 = OPEN</td><td>08 = BUS-2EASY device check in progress</td></tr><tr><td>02 = Stop then "OPEN"</td><td>09 = Preflashing then "OPEN"</td></tr><tr><td>03 = Stop then "CLOSE"</td><td>10 = Preflashing then "CLOSE"</td></tr><tr><td>04 = In "PAUSE"</td><td>11 = OPEN in Partial Opening</td></tr><tr><td>05 = During opening stage</td><td>12 = in PAUSE Partial Opening</td></tr><tr><td>06 = During closing stage</td><td></td></tr></table>	00 = CLOSED	07 = FAIL SAFE in progress	01 = OPEN	08 = BUS-2EASY device check in progress	02 = Stop then "OPEN"	09 = Preflashing then "OPEN"	03 = Stop then "CLOSE"	10 = Preflashing then "CLOSE"	04 = In "PAUSE"	11 = OPEN in Partial Opening	05 = During opening stage	12 = in PAUSE Partial Opening	06 = During closing stage		
00 = CLOSED	07 = FAIL SAFE in progress															
01 = OPEN	08 = BUS-2EASY device check in progress															
02 = Stop then "OPEN"	09 = Preflashing then "OPEN"															
03 = Stop then "CLOSE"	10 = Preflashing then "CLOSE"															
04 = In "PAUSE"	11 = OPEN in Partial Opening															
05 = During opening stage	12 = in PAUSE Partial Opening															
06 = During closing stage																

6 STORING RADIO CODE

The electronic device is equipped with an integrated dual channel decoding system (DS, SLH, LC/RC) known as OMNIDEC. This system can use an additional receiving module (Fig. 5 re. ①) and radiocontrols

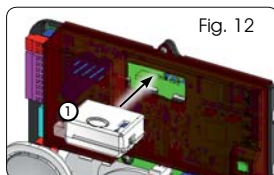


Fig. 12

of the same frequency to store both total opening (OPEN A) and partial opening (OPEN B) of the automated system.

The 3 types of radio coding (DS, SLH, LC/RC) can co-exist at the same time on the two channels. It is possible to enter up to 256 radio codes subdivided between OPEN A and OPEN B.

To use the different coding systems on the same channel, finish the learning process for each system and then repeat the procedure for another.

6.1 STORING DS RADIOCONTROLS

It is possible to store a maximum of 256 codes divided between OPEN A and OPEN B/CLOSE.

- On the DS radiocontrol, select the required ON - OFF combination of the 12 dip-switches.
- Press and hold down push-button + (SW3) or - (SW2) and then press the SETUP push-button (SW4), to store total opening (OPEN A) or partial opening (OPEN B/CLOSE) respectively. The corresponding led will begin to flash slowly for 5 secs.
- Release both push-buttons.
- Within this 5 second period, press the required push-button on the radiocontrol.
- The corresponding led will come on with a fixed light for 1 second and then go off to indicate the setting has been stored.
- To add other codes other than the one you have recently stored, repeat the procedure from point 1.

7. To use radiocontrols with the same code, copy the ON - OFF combination to the other radiocontrols without the need to change anything on the control board.

6.2 STORING SLH RADIOCONTROLS

 ***It is possible to store a maximum of 256 codes divided between OPEN A and OPEN B/CLOSE.***

1. On the SLH radiocontrol, press and hold down P1 and P2 simultaneously.
2. The radiocontrol led will begin to flash.
3. Release both push-buttons.
4. Press and hold down push-button + (SW3) or - (SW2) and then press the SETUP push-button (SW4), to store total opening (OPEN A) or partial opening (OPEN B/CLOSE) respectively. The corresponding led will begin to flash slowly for 5 secs.
5. Release both push-buttons.
6. Within this 5 sec. period while the radiocontrol led is still flashing, press and hold down the required push-button on the radiocontrol (the radiocontrol led will come on with a fixed light).
7. The corresponding led will come on with a fixed light for 2 seconds and then go off to indicate the setting has been stored.
8. Release the radiocontrol push-button.
9. Press the stored radiocontrol push-button twice in quick succession.

 ***The automated system will carry out an opening control. Ensure the automated device is free of any obstacle by people or things.***

To enable other radiocontrols with the same system code, transfer the system code of the stored radiocontrol push-button to the corresponding push-button of the radiocontrols to be added, following the procedure below:

- a. On the stored radiocontrol, press and hold down P1 and P2 simultaneously.
- b. The radiocontrol led will begin to flash.
- c. Release both push-buttons.
- d. Press the stored push-button and hold down (the radiocontrol led will come on with a fixed light).
- e. Bring the radiocontrols closer to one another, press and hold down the corresponding push-button of the radiocontrol to be added, releasing only once the led on the radiocontrol has issued a double flash to indicate that it has been stored.
- f. Press the stored radiocontrol push-button twice in quick succession.

 ***The automated system will carry out an opening control. Ensure the automated device is free of any obstacle by people or things.***

6.3 STORING LC/RC RADIOCONTROLS

(ONLY FOR SOME MARKETS)

 ***It is possible to store a maximum of 256 codes divided between OPEN A and OPEN B/CLOSE.***

 ***Use the LC/RC remote controls only with a 433 MHz receiving module.***

1. Press and hold down push-button + (SW3) or - (SW2) and then press the SETUP push-button (SW4), to store total opening (OPEN A) or partial opening (OPEN B/CLOSE) respectively. The corresponding led will begin to flash slowly for 5 secs.
2. Release the push-button.
3. Within this 5 second period, press the required push-button on the LC/RC remote control.
4. The led will come on with a fixed light for 1 second, indicating that the unit has been stored, and then resume flashing for an additional 5 seconds during which time another radiocontrol may be stored.
5. Once the 5 seconds are up, the led will go off to indicate that the procedure is complete.

6.3.1 STORING LC/RC RADIOCONTROLS REMOTELY

With LC/RC radiocontrols, other radiocontrols may be stored in remote mode, i.e. without operating directly on the board, using a previously stored radiocontrol.

1. Obtain a radiocontrol already stored on one of its 2 channels (OPEN A or OPEN B/CLOSE).
2. Press and hold down push-buttons P1 and P2 simultaneously until both leds flash slowly for 5 secs.
3. Within 5 seconds, push the previously stored radiocontrol push-button to activate the learning stage on the selected channel.
4. The led on the board corresponding to the channel being stored flashes for 5 seconds, within which time the code of another radiocontrol must be sent.
5. The led will come on with a fixed light for 1 second, indicating that the unit has been stored, and then resume flashing for an additional 5 seconds during which time other radiocontrols may be stored and then it will go off.

6.4 RADIOCONTROLS DELETION PROCEDURE

To delete **ALL** the codes of the radiocontrols entered, simply press down together the two push-buttons + (SW3) and - (SW2), then press the SETUP push-button (SW4) for 1 second, and hold the first two push-buttons pressed for 10 secs.

- The 2 leds RADIO1 and RADIO2 will flash quickly for 10 secs.
- The 2 leds will come on with a fixed light for 2 seconds and then go off (deletion carried out).
- Release both push-buttons.

 ***This operation is NOT reversible. All the codes of the radiocontrols stored as OPEN A and as OPEN B/CLOSE will be deleted.***

7 OPERATIONAL START-UP

7.1 CHECKING THE LEDS

After making all the connections and powering the board, check the status of the leds against the status of the inputs in the following table (in Fig. 6, the condition where the automated system is closed in stand-by). These LEDs indicate the status of the board inputs and are of considerable importance to automated system movement:

Note that:  **LED ON** = contact closed  **LED OFF** = contact open

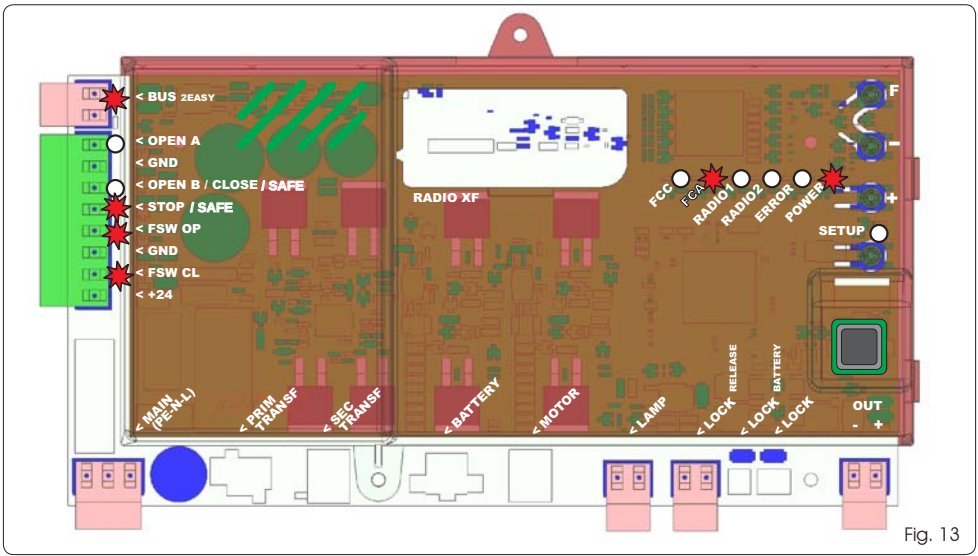


Fig. 13

 The inputs STOP, FSW CL, FSW OP are safety inputs with a N.C. (Normally Closed) contact and the corresponding leds must therefore be ON; where the automated system is in stand-by, they will go off when the connected device is engaged.

 The inputs OPEN A, OPEN B/CLOSE are inputs with a N.O. (Normally Open) contact and the corresponding leds must therefore be OFF; where the automated system is in stand-by, they will come on when the connected device is engaged.

 **ERROR led flashing indicates alarm in progress (situation does not impair gate operation)**

 **ERROR led on with a fixed light indicates an error in progress (situation that locks operation until the cause of the error has been removed)**

Leds FCA and FCC represent the N.C. contacts of the limit switch built into the board that open when engaged and turn off the associated led:

Automated system CLOSED	FCA 
	FCC  FCC engaged
Automated system OPEN	FCA  FCA engaged
	FCC 

Tab. 3 - Description of POWER led

Fixed ON	Mains-powered
Flashing	Battery-powered
Off	Board off

7.2 POSITIONING LIMIT SWITCHES

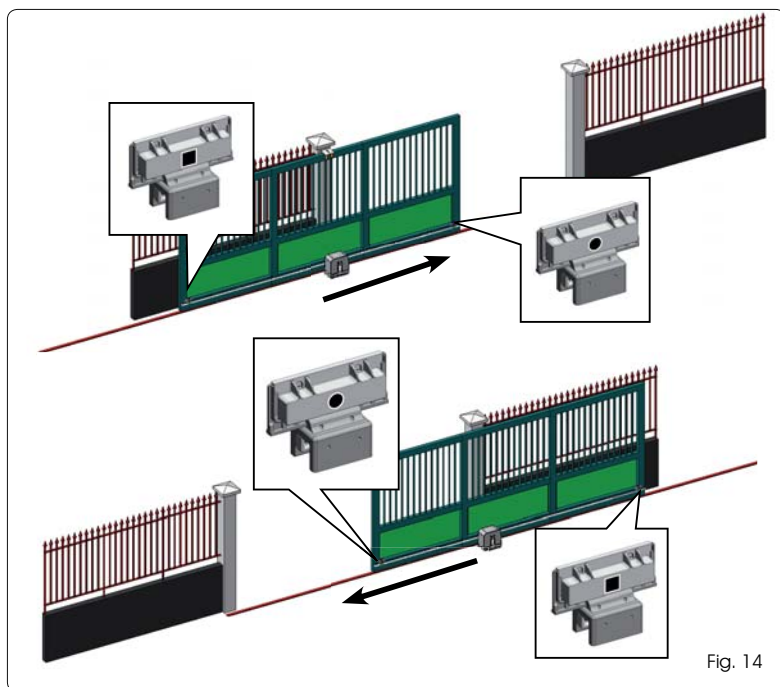
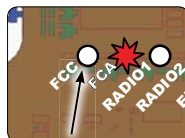
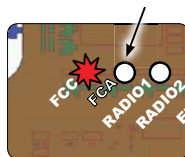
To ensure correct positioning of the limit switch magnets, the control unit must be installed and correctly connected with all control and safety accessories.

The operator is equipped with a magnetic limit switch sensor built directly into the electronic control board. The gate is stopped, during opening or closing, when the polarised magnet secured to the upper part of the rack activates the sensor.

1. Check that the operator is set for manual operating mode as described in the instructions for the operator.
2. Move the gate to **opening** position manually, leaving 40 mm from the limit switch mechanical stop (see Fig. 06).
3. Slide the magnet with the **CIRCLE** on the rack in the direction of the motor. As soon as the led for the FCA limit switch on the board goes off, secure with the appropriate screws.
4. Move the gate to **closing** position manually, leaving 40 mm from the limit switch mechanical stop.
5. Slide the magnet with the **SQUARE** on the rack in the direction of the motor. As soon as the led for the FCC limit switch on the board goes off, secure with the appropriate screws.
6. Check that the relevant limit switch led goes off correctly at the end of the opening and closing movement and make the necessary changes to the position of the limit switch magnet position if necessary.



To avoid damage to the operator and/or interruptions in the operation of the automated system, approximately 40 mm must be left from the mechanical limit switch stops.



For correct operation of the operator, the magnet with the **CIRCLE** must be used as an **OPENING** limit switch and the magnet with the **SQUARE** must be used as a **CLOSING** limit switch.

(SEE FIG. 07)

Fig. 14

7.3 SETUP

 **When the board is powered when no SETUP has ever been carried out, or if the board requires it, the code 50 will flash on the display together with the SETUP led to indicate that SETUP must be carried out.**

 **To ensure the effective outcome of the SETUP procedure, carefully check the correct polarity of the magnetic limit switches as described in the previous paragraph.**

Proceed as follows to carry out the SETUP:

1. **Move the gate leaf to mid-way through its travel (very important to the success of the SETUP) and check that both leds FCA and FCC are on. Otherwise, the board will indicate error 12 (see tab. 5)**
2. Press and hold the SETUP (SW4) push-button until the gate begins to move slowly and stops when it reaches the limit switch. If the limit switch reached is the closing switch (with the **SQUARE**) the device stores that point as the **closing** stop, conversely if the limit switch is the opening switch (with the **CIRCLE**) the device stores that point as the **opening** stop. During that stage, **51** flashes on the display
3. The gate automatically begins to move slowly in the opposite direction and then stops when it reaches the limit switch. If the limit switch reached is the opening switch (with the **SQUARE**) the device stores that point as the **opening** stop, conversely if the limit switch is the closing switch (with the **SQUARE**) the device stores that point as the **closing** stop. During that stage, **53** flashes on the display
4. According to the final limit switch reached, the device takes up a status of closed (**00**) or open (**01**). In the second case, issue an OPEN pulse to close the gate.

8 TESTING THE AUTOMATED SYSTEM

After installation and programming, check the system is operating correctly. Above all, check that the safety devices operate correctly and ensure that the current safety regulations are met.

9 ALARM AND ERROR SIGNALS

If **alarms** occur (conditions that do not impair the operation of the gate) or **errors** (conditions that lock the gate operation) the display shows a number relating to the current condition.

 **The ALARM or ERROR signals disappear at the next cycle only if the triggering cause is removed.**

9.1 ALARMS

 **When an ALARM occurs, the ERROR led begins to flash and a number relating to the current fault appears on the display when the + and - keys are pressed.**

Tab. 4 indicates all the alarms that may be shown on the display.

Tab. 4 - Alarms

22	Limited MOTOR current
24	LAMP output short-circuited
27	Obstacle detection (visible for 10 secs)
30	XF radio code memory-module full (visible for 10 secs)
40	Service request
46	Forced default programming reset

9.2 ERRORS

 **When an ERROR occurs, led DL20 comes on with a fixed light and a number relating to the current fault appears on the display when the + and - keys are pressed.**

Tab. 5 indicates all the errors that may be shown on the display.

Tab. 5 - Errors

01	Board faulty
03	Motor faulty
06	Motor lock locked closed (check the motor lock and replace if necessary)
07	Gate too heavy or too much friction (try to increase motor power)
08	BUS-2EASY device error (e.g. same address on two photocell pairs; check address)
10	Both limit switches with the same polarity
12	Limit switch engaged at the beginning of SETUP
15	Time-out finished

10 FUNCTION LOGICS

 **The effects on the other active pulse inputs are shown in brackets**

 **The CLOSE can be activated at the OPEN B input from the 2nd programming level**

LOGIC "E"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens the gate	opens the gate partially	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	locks operation (1)	locks operation	recloses the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (OPEN locks - stores CLOSE)
OPEN	recloses the gate (1)	recloses the gate	recloses the gate	no effect (OPEN/CLOSE inhibited)	no effect	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (OPEN locks - stores CLOSE)
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN locks - stores CLOSE)

LOGIC "EP"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens the gate	opens the gate partially	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	locks operation (1)	locks operation	recloses the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (OPEN locks - stores CLOSE)
OPEN	recloses the gate (1)	recloses the gate	recloses the gate	no effect (OPEN/CLOSE inhibited)	no effect	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)
DURING CLOSING	locks operation	locks operation	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (OPEN locks - stores CLOSE)
LOCKED	resumes motion in reverse direction. After STOP, always closes	resumes motion in reverse direction. After STOP, always closes	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN locks - stores CLOSE)

LOGIC "A"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect (1)	no effect	recloses the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (stores CLOSE)
OPEN IN PAUSE	reloads pause time (1)	reloads pause time B	recloses the gate	locks operation	no effect	reloads pause time (CLOSE inhibited)	reloads pause time (CLOSE inhibited)
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores CLOSE)
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC "A1"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect (1)	no effect	recloses the gate	locks operation	see 2nd level prog.	continues to open and recloses immediately	locks and opens on disengagement (stores CLOSE)
OPEN IN PAUSE	reloads pause time (1)	reloads pause time B	recloses the gate	locks operation	no effect	locks and immediately closes on disengagement	reloads pause time (CLOSE inhibited)
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores CLOSE)
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

(1) During the partial opening cycle, an OPEN A pulse causes total opening

LOGIC "AP"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	locks operation (1)	locks operation	recloses the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (OPEN locks - stores CLOSE)
OPEN IN PAUSE	locks operation (1)	locks operation	recloses the gate	locks operation	no effect	reloads pause time (CLOSE inhibited)	reloads pause time (CLOSE inhibited)
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (OPEN locks - stores CLOSE)
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC "A1" (2)	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect (1)	no effect	recloses the gate	locks operation	reverses to closing (1)	no effect	locks and opens on disengagement (stores CLOSE)
OPEN IN PAUSE	reloads pause time (1)	reloads pause time	recloses the gate	locks operation	no effect	reloads pause time (CLOSE inhibited)	reloads pause time (CLOSE inhibited)
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores CLOSE)
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC "S"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	reverses to closing (1)	reverses to closing (1)	recloses the gate	locks operation	see 2nd level prog.	continues to open and recloses immediately	locks and opens on disengagement (stores CLOSE)
OPEN IN PAUSE	recloses the gate (1)	recloses the gate	recloses the gate	locks operation	no effect	locks and immediately closes on disengagement	locks and closes on disengagement
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level programming) and closes immediately at the end	locks and on disengagement opens and at the end immediately closes
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC "SP"	PULSES						
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	locks operation (1)	locks operation	recloses the gate	locks operation	see 2nd level prog.	continues to open and recloses immediately	locks and on disengagement opens and at the end immediately closes (OPEN locks - stores CLOSE)
OPEN IN PAUSE	recloses the gate (1)	recloses the gate	recloses the gate	locks operation	no effect	locks and immediately closes on disengagement	locks and immediately closes on disengagement
DURING CLOSING	locks operation	locks operation	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores CLOSE)
LOCKED	resumes motion in reverse direction. After STOP, always closes	resumes motion in reverse direction. After STOP, always closes	recloses the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

(1) During the partial opening cycle, an OPEN A pulse causes total opening

LOGIC 'SA'		PULSES					
STATUS OF AUTOMATED SYSTEM	OPEN A	OPEN B	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens and recloses after the pause time	Opens the gate partially and closes it after pause time B	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect (1)	no effect	recloses the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (stores CLOSE)
OPEN IN PAUSE	recloses the gate (1)	recloses the gate	recloses the gate	locks operation	no effect	reloads pause time (CLOSE inhibited)	reloads pause time (CLOSE inhibited)
DURING CLOSING	reopens the gate	reopens the gate	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores CLOSE)
LOCKED	closes the gate	closes the gate	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC 'B'		PULSES					
STATUS OF AUTOMATED SYSTEM	OPEN A	/	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens the gate	/	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect	/	closes the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (stores OPEN/CLOSE)
OPEN	no effect	/	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)
DURING CLOSING	opens the gate	/	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores OPEN/CLOSE)
LOCKED	opens the gate	/	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC 'bC'		PULSES DURING OPENING/COMMANDS MAINTAINED DURING CLOSING			PULSES		
STATUS OF AUTOMATED SYSTEM	OPEN A	/	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens the gate	/	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect	/	closes the gate	locks operation	see 2nd level prog.	no effect	locks and opens on disengagement (stores OPEN/CLOSE)
OPEN	no effect	/	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)
DURING CLOSING	opens the gate	/	no effect	locks operation	no effect	reverses to opening (see 2nd level prog.)	locks and opens on disengagement (stores OPEN/CLOSE)
LOCKED	opens the gate	/	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

LOGIC 'C'		COMMANDS MAINTAINED		PULSES			
STATUS OF AUTOMATED SYSTEM	OPEN A	/	CLOSE	STOP	FSW OP	FSW CL	FSW CL/OP
CLOSED	opens the gate	/	no effect	no effect (OPEN inhibited)	no effect (OPEN inhibited)	no effect	no effect (OPEN inhibited)
DURING OPENING	no effect	/	closes the gate	locks operation	see 2nd level prog.	no effect	locks and on disengagement 2nd level prog.
OPEN	no effect	/	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)
DURING CLOSING	opens the gate	/	no effect	locks operation	no effect	locks operation	locks operation
LOCKED	opens the gate	/	closes the gate	no effect (OPEN/CLOSE inhibited)	no effect (OPEN inhibited)	no effect (CLOSE inhibited)	no effect (OPEN/CLOSE inhibited)

(1) During the partial opening cycle, an OPEN A pulse causes total opening

(2) When turned on, the board checks the inputs and if an OPEN A or B command is active, it opens the leaf or gate. Otherwise it closes it.

Le descrizioni e le illustrazioni del presente manuale non sono impegnative. La FAAC si riserva il diritto, lasciando inalterate le caratteristiche essenziali dell'apparecchiatura, di apportare in qualunque momento e senza impegnarsi ad aggiornare la presente pubblicazione, le modifiche che essa ritiene convenienti per miglioramenti tecnici o per qualsiasi altra esigenza di carattere costruttivo o commerciale.

The descriptions and illustrations contained in the present manual are not binding. FAAC reserves the right, whilst leaving the main features of the equipments unaltered, to undertake any modifications it holds necessary for either technical or commercial reasons, at any time and without revising the present publication.

Les descriptions et les illustrations du présent manuel sont fournies à titre indicatif. FAAC se réserve le droit d'apporter à tout moment les modifications qu'elle jugera utiles sur ce produit tout en conservant les caractéristiques essentielles, sans devoir pour autant mettre à jour cette publication.

Die Beschreibungen und Abbildungen in vorliegendem Handbuch sind unverbindlich. FAAC behält sich das Recht vor, ohne die wesentlichen Eigenschaften dieses Gerätes zu verändern und ohne Verbindlichkeiten in Bezug auf die Neufassung der vorliegenden Anleitungen, technisch bzw. konstruktiv/kommerziell bedingte Verbesserungen vorzunehmen.

Las descripciones y las ilustraciones de este manual no comportan compromiso alguno. FAAC se reserva el derecho, dejando inmutadas las características esenciales de los aparatos, de aportar, en cualquier momento y sin comprometerse a poner al día la presente publicación, todas las modificaciones que considere oportunas para el perfeccionamiento técnico o para cualquier otro tipo de exigencia de carácter constructivo o comercial.

De beschrijvingen in deze handleiding zijn niet bindend. FAAC behoudt zich het recht voor op elk willekeurig moment de veranderingen aan te brengen die het bedrijf nuttig acht met het oog op technische verbeteringen of alle mogelijke andere productie- of commerciële eisen, waarbij de fundamentele eigenschappen van de apparaat gehandhaafd blijven, zonder zich daardoor te verplichten deze publicatie bij te werken.



FAAC

FAAC S.p.A.

Via Calari, 10
40069 Zola Predosa (BO) - ITALIA
Tel. 0039.051.61724 - Fax. 0039.051.758518
www.faac.it
www.faacgroup.com

